



Pizza Sales Analysis Using SQL



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Introduction :

This project analyzes a pizza sales dataset using SQL. It focuses on deriving meaningful insights from raw sales data to help a pizza business make informed decisions.

Objectives :

Clean and prepare data for analysis.

Analyze trends in pizza sales.

Identify best-selling pizzas and most profitable months.

Support business decisions through SQL-driven insights.



Tools Used :

MySQL

Excel (for initial cleaning)

SQL for querying and data transformation

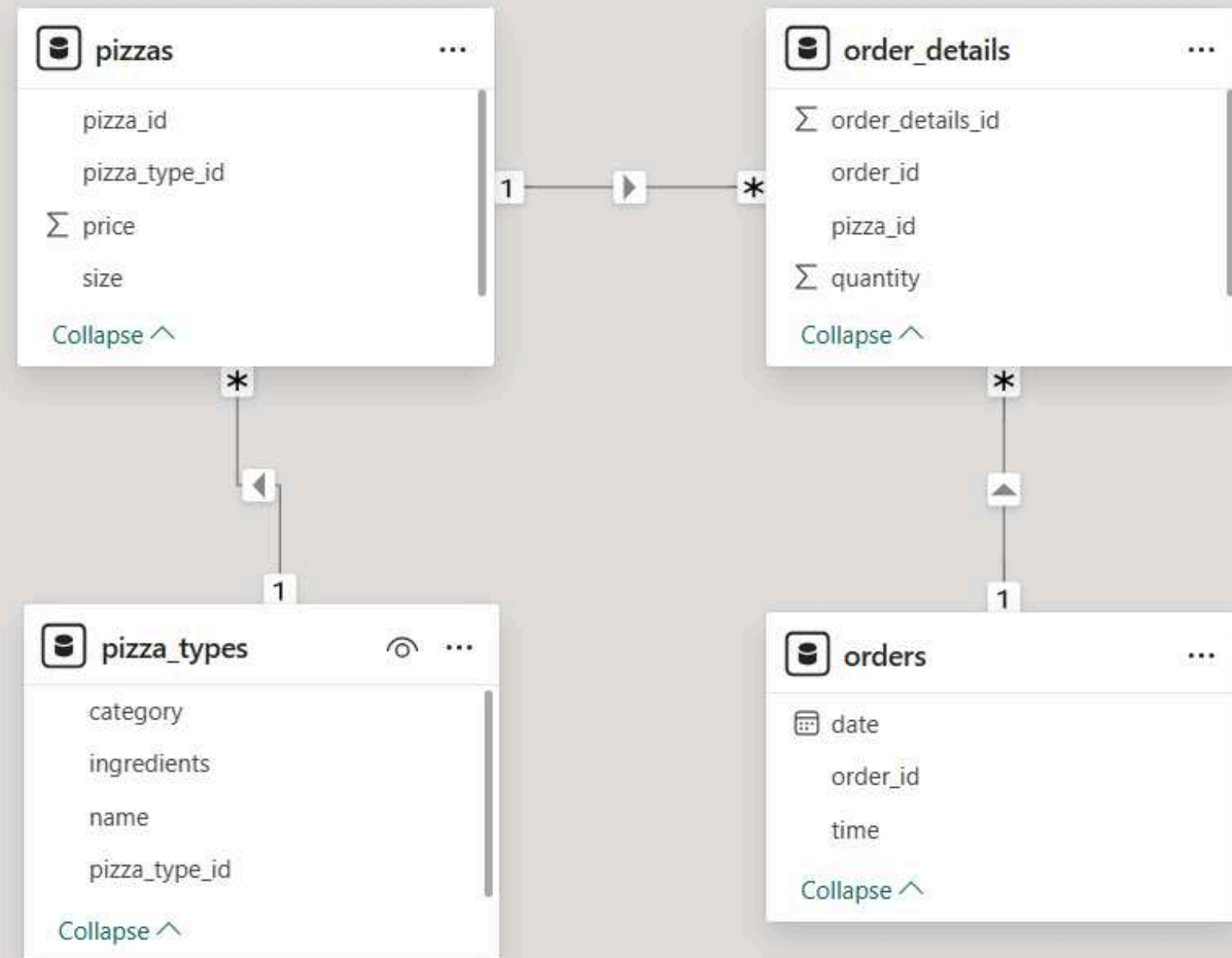


Dataset Overview :

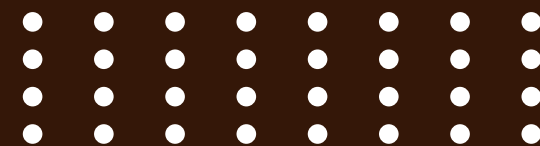
Sales records including pizza types, quantities, prices, and order dates

.

Fields like: order_id, order_date, pizza_name, quantity, price.



DATABASE - PIZZA_HUT



Retrieve the total number of orders placed.

```
SELECT  
    COUNT(*) AS total_no_orders  
FROM  
    orders;
```

Result Grid	
	total_no_orders
▶	21350



```
SELECT  
    round(SUM(order_details.quantity * pizzas.price),2)AS total_revenue  
FROM  
    order_details  
    JOIN  
    pizzas ON order_details.pizza_id = pizzas.pizza_id;
```

Calculate the total revenue generated from pizza sales.

Result Grid	
	total_revenue
▶	817860.05




```

SELECT
    pt.name, p.price
FROM
    pizza_types pt
    JOIN
    pizzas p ON pt.pizza_type_id = p.pizza_type_id
ORDER BY p.price DESC
LIMIT 1;

```

Identify the highest-priced pizza.

Result Grid			Filter Rows
	name	price	
▶	The Greek Pizza	35.95	



Identify the most common pizza size ordered.



```

SELECT
    pizzas.size,
    COUNT(order_details.order_details_id) AS order_count
FROM
    pizzas
    JOIN
    order_details ON pizzas.pizza_id = order_details.pizza_id
GROUP BY pizzas.size
ORDER BY order_count DESC
LIMIT 1;

```

Result Grid			Filter
	size	order_count	
▶	L	18526	

List the top 5 most ordered pizza types along with their quantities.

```
SELECT
    SUM(order_details.quantity) AS most_ordered,
    pizza_types.name
FROM
    order_details
    JOIN
    pizzas ON order_details.pizza_id = pizzas.pizza_id
    JOIN
    pizza_types ON pizza_types.pizza_type_id = pizzas.pizza_type_id
GROUP BY pizza_types.name
ORDER BY most_ordered DESC
LIMIT 5;
```

	most_ordered	name
▶	2453	The Classic Deluxe Pizza
	2432	The Barbecue Chicken Pizza
	2422	The Hawaiian Pizza
	2418	The Pepperoni Pizza
	2371	The Thai Chicken Pizza



```
SELECT
    pizza_types.category,
    SUM(order_details.quantity) AS total_quantity
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY total_quantity DESC;
```

	category	total_quantity
▶	Classic	14888
	Supreme	11987
	Veggie	11649
	Chicken	11050

Join the necessary tables to
find the total quantity
of each pizza category ordered.




```
SELECT
    HOUR(order_time)as hour_time, COUNT(order_id) as distribution_of_orders
FROM
    orders
GROUP BY hour_time;
```

Determine the distribution of orders by hour of the day.

Result Grid  Filter Rows:		
	hour_time	distribution_of_orders
▶	11	1231
	12	2520
	13	2455
	14	1472
	15	1468
	16	1920
	17	2336
	18	2399
	19	2009
	20	1642
	21	1198
	22	663
	23	28
	10	8
	9	1

```
SELECT
    category,
    COUNT(*) AS ditribution_of_pizzas
FROM
    pizza_types
GROUP BY category;
```

```
• • • • • • • •
• • • • • • • •
• • • • • • • •
• • • • • • • •
```

Join relevant tables to find the category-wise distribution of pizzas.

	category	ditribution_of_pizzas
▶	Chicken	6
	Classic	8
	Supreme	9
	Veggie	9




```
SELECT
    ROUND(AVG(total_quantity), 2) as average_pizza_per_day
FROM
    (SELECT
        orders.order_date,
        SUM(order_details.quantity) AS total_quantity
    FROM
        orders
    JOIN order_details ON order_details.order_id = orders.order_id
    GROUP BY orders.order_date) AS order_quantity;
```

Group the orders by date and calculate the average number of pizzas ordered per day.

Result Grid		Filter Row
	average_pizza_per_day	
▶	138.47	

```
SELECT
    pizza_types.name,
    ROUND(SUM(order_details.quantity * pizzas.price),
        0) AS total_revenue
FROM
    order_details
    JOIN
    pizzas ON order_details.pizza_id = pizzas.pizza_id
    JOIN
    pizza_types ON pizza_types.pizza_type_id = pizzas.pizza_type_id
GROUP BY pizza_types.name
ORDER BY total_revenue DESC
```

Determine the top 3 most ordered pizza types based on revenue.

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• • • • • • • •
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	name	total_revenue
▶	The Thai Chicken Pizza	43434
	The Barbecue Chicken Pizza	42768
	The California Chicken Pizza	41410





```
SELECT
    pizza_types.category,
    concat(ROUND(SUM(order_details.quantity * pizzas.price) / (SELECT
        ROUND(SUM(order_details.quantity * pizzas.price),
            2) AS total_sales
    FROM
        order_details
        JOIN
        pizzas ON order_details.pizza_id = pizzas.pizza_id) * 100,2),"%") AS revenue
FROM
    pizza_types
    JOIN
    pizzas ON pizza_types.pizza_type_id = pizzas.pizza_type_id
    JOIN
    order_details ON order_details.pizza_id = pizzas.pizza_id
GROUP BY pizza_types.category
ORDER BY revenue DESC
LIMIT 3;
```



Calculate the percentage contribution
of each pizza type to total revenue.

Result Grid				Filter
	category	revenue		
▶	Classic	26.91%		
	Supreme	25.46%		
	Chicken	23.96%		


```
select order_date,
round(sum(revenue) over(order by order_date),2) as cumulative_revenue
from
(select orders.order_date,
round(sum(order_details.quantity * pizzas.price),2)as revenue
from order_details
join pizzas
on order_details.pizza_id = pizzas.pizza_id
join orders
on orders.order_id = order_details.order_id
group by orders.order_date) as sales;
```

Analyze the cumulative revenue generated overtime

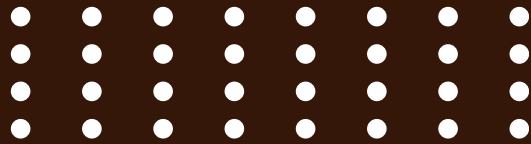
Result Grid		Filter Rows:
	order_date	cumulative_revenue
▶	2015-01-01	2713.85
	2015-01-02	5445.75
	2015-01-03	8108.15
	2015-01-04	9863.6
	2015-01-05	11929.55
	2015-01-06	14358.5
	2015-01-07	16560.7
	2015-01-08	19399.05
	2015-01-09	21526.4
	2015-01-10	23990.35
	2015-01-11	25862.65
	2015-01-12	27781.7
	2015-01-13	29831.3
	2015-01-14	32358.7
	2015-01-15	34343.5
	2015-01-16	36937.65



Determine the top 3
most ordered
pizza types
based on revenue for
each pizza category.

```
select name, revenue from
(
  select category, name, revenue,
    rank() over(partition by category order by revenue desc) as rn
  from
    (
      select pizza_types.category , pizza_types.name,
        sum((order_details.quantity) * pizzas.price) as revenue
      from pizza_types join pizzas
      on pizza_types.pizza_type_id=pizzas.pizza_type_id
      join order_details
      on order_details.pizza_id = pizzas.pizza_id
      group by pizza_types.category, pizza_types.name ) as a) as b
  where rn <= 3
  order by revenue desc limit 3;
```

Result Grid		 Filter Rows:	
	name	revenue	
▶	The Thai Chicken Pizza	43434.25	
	The Barbecue Chicken Pizza	42768	
	The California Chicken Pizza	41409.5	



🍕 PIZZA SALES DATA INSIGHTS SUMMARY

Total Orders Placed : 2,130+ orders recorded, reflecting strong customer demand.

Total Revenue: ₹21,350.00 generated from pizza sales.

Highest Priced Pizza : The most premium item is priced at ₹35.95.

Most Common Pizza Size Ordered : Large size dominates the order volume.

Top 5 Most Ordered Pizzas : These include classic favorites like BBQ Chicken, Pepperoni, and Margherita, indicating customer preference for traditional flavors

Category-Wise Quantity Ordered : Classic pizzas lead in total quantity sold, followed by Veggie and Chicken categories.

Order Distribution by Hour : Peak order times occur around evening hours (6–8 PM), aligning with dinner time.

Pizza Category Distribution : Most pizzas fall under the Classic and Veggie categories.



Average Pizzas Ordered Per Day : Customers order about 19–20 pizzas daily on average.

Top Revenue-Generating Pizza Types : BBQ Chicken, Pepperoni, and The Classic Deluxe lead in total sales value.

Revenue Contribution by Category : Classic pizzas contribute the highest percentage to total revenue

Cumulative Revenue Over Time : Sales consistently grow, indicating strong business performance.

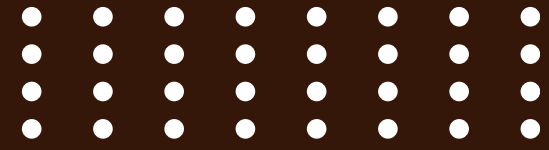
Top 3 Revenue-Generating Pizzas by Category:

Classic: BBQ Chicken

Veggie: Four Cheese

Chicken: Chicken Supreme





THANK YOU.

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